Adaptive Memory Dynamics - Stage Two Domain-Shift Results

Status: preliminary exploratory software experiment, not peer reviewed.

This report is generated directly from the Stage Two CSV and manifest. Historical Stage One findings are preserved; this experiment addresses a **different question** (same labels, changed image domain).

Configuration and evidence

- Dataset: scikit-learn load_digits, 8x8 handwritten digits; no MNIST.

- Seeds: 7, 21, 42, 84, 101 (5 matched seeds per condition).

- Conditions: 7 (persistent; passive decay; usage-reinforced decay).

- Domains: rotate90, permuted_pixels.

- Training: 8 epochs for each of A and B; 32 hidden neurons; SGD learning rate 0.08.

- Complete condition runs: 70; epoch metric rows: 1120.

- Source commit: `339732702b19a0dc0ff41d716dc3a9c20301cb61`.

- Input data, labels, splits, initialization seeds, and minibatch order are matched within each domain. Transformed test samples are held out from training, but the seeds/rates are exploratory rather than a preregistered independent confirmatory test.

Held-out accuracy and forgetting, by configuration

Values below are the **mean +/- sample standard deviation across seeds** when at least two seeds are available. Higher A-after-B and B-after-B accuracy is preferable. Lower zero-clipped forgetting is preferable. These are proportions, not percentage points.

Domain B: `rotate90`

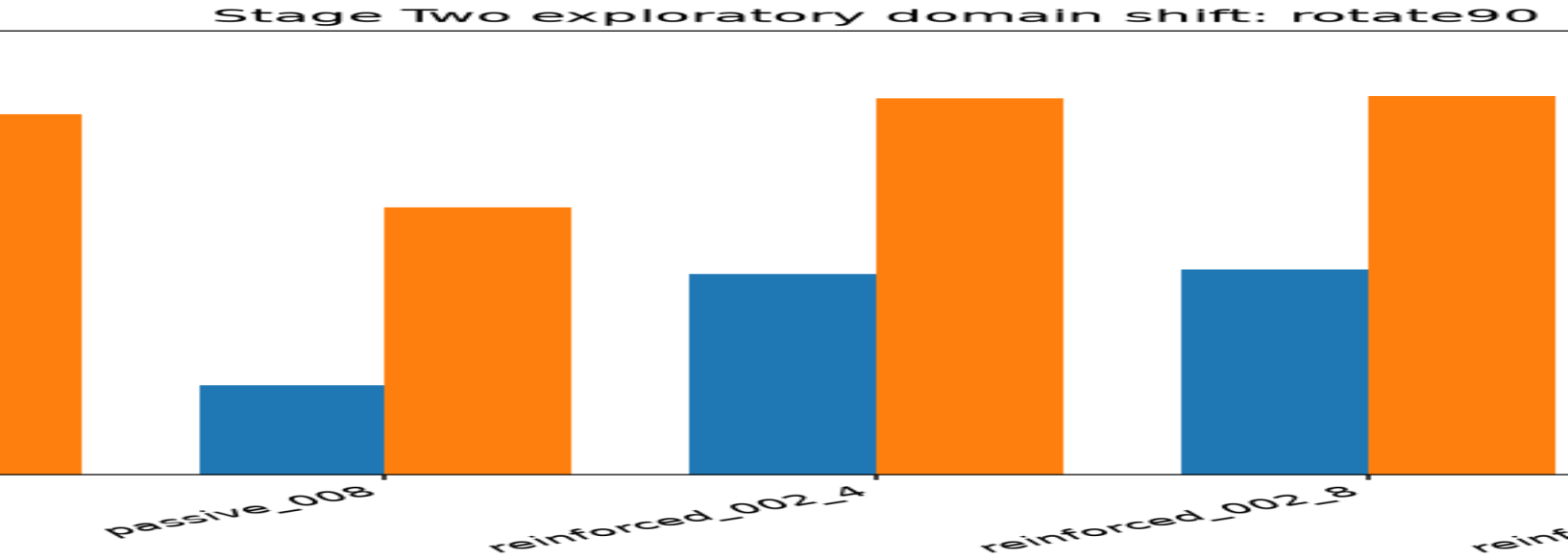
Condition	Task A after B	Task B after A	A forgetting	Mean final accuracy
persistent	0.587 +/- 0.065	0.879 +/- 0.013	0.220 +/- 0.077	0.733 +/- 0.031
passive_002	0.362 +/- 0.032	0.812 +/- 0.025	0.426 +/- 0.064	0.587 +/- 0.027
passive_008	0.201 +/- 0.025	0.603 +/- 0.048	0.512 +/- 0.055	0.402 +/- 0.026
reinforced_002_4	0.452 +/- 0.050	0.848 +/- 0.014	0.340 +/- 0.085	0.650 +/- 0.028
reinforced_002_8	0.463 +/- 0.055	0.852 +/- 0.014	0.330 +/- 0.085	0.658 +/- 0.030
reinforced_008_4	0.344 +/- 0.033	0.772 +/- 0.034	0.427 +/- 0.057	0.558 +/- 0.033
reinforced_008_8	0.400 +/- 0.045	0.801 +/- 0.026	0.384 +/- 0.083	0.600 +/- 0.033

Paired difference in Task A retention vs persistent (mean, exploratory 95% percentile bootstrap interval):

- `passive_002`: -0.225 ([-0.251, -0.200]; n=5).

- `passive_008`: -0.386 ([-0.424, -0.341]; n=5).

- `reinforced_002_4`: -0.135 ([-0.160, -0.114]; n=5).
- `reinforced_002_8`: -0.124 ([-0.149, -0.103]; n=5).
- `reinforced_008_4`: -0.244 ([-0.272, -0.213]; n=5).
- `reinforced_008_8`: -0.188 ([-0.216, -0.166]; n=5).

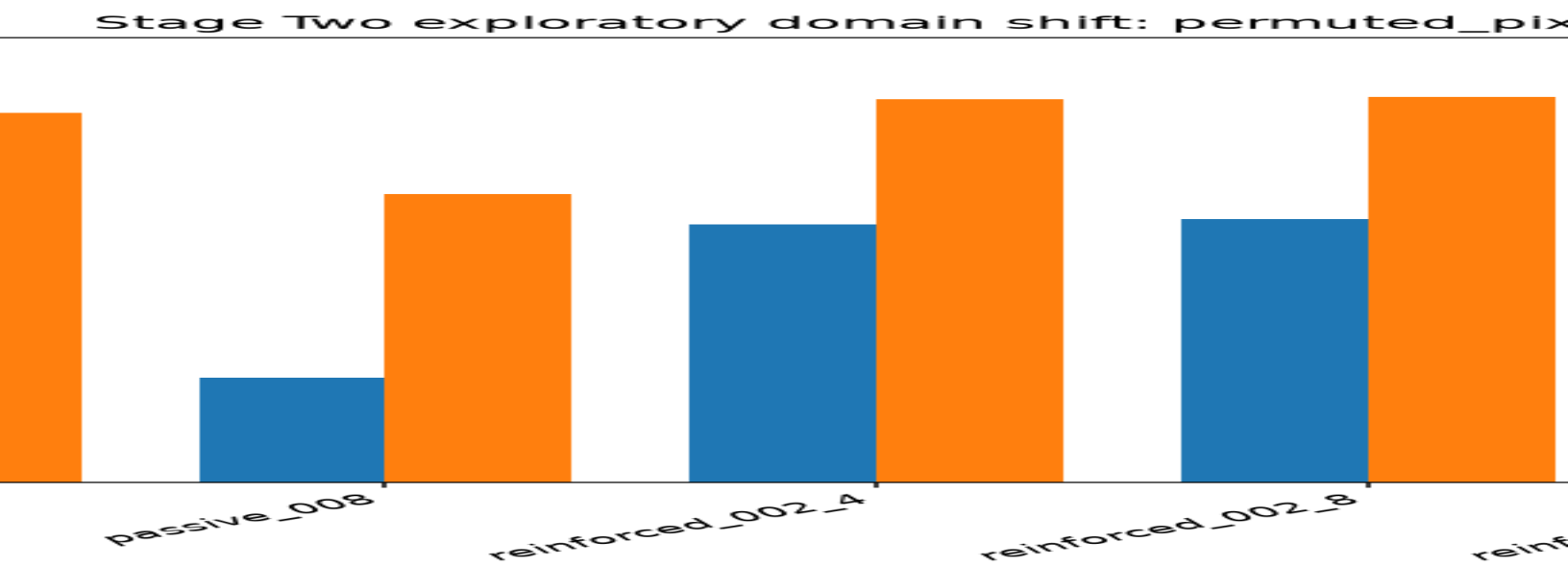


Domain B: `permuted_pixels`

Condition	Task A after B	Task B after A	A forgetting	Mean final accuracy
persistent	0.607 +/- 0.066	0.885 +/- 0.020	0.201 +/- 0.098	0.746 +/- 0.028
passive_002	0.499 +/- 0.100	0.830 +/- 0.021	0.289 +/- 0.131	0.665 +/- 0.046
passive_008	0.235 +/- 0.058	0.649 +/- 0.032	0.479 +/- 0.117	0.442 +/- 0.023
reinforced_002_4	0.580 +/- 0.081	0.861 +/- 0.014	0.212 +/- 0.120	0.720 +/- 0.040
reinforced_002_8	0.591 +/- 0.080	0.866 +/- 0.011	0.202 +/- 0.119	0.729 +/- 0.040
reinforced_008_4	0.491 +/- 0.103	0.806 +/- 0.032	0.279 +/- 0.136	0.649 +/- 0.043
reinforced_008_8	0.536 +/- 0.096	0.831 +/- 0.023	0.248 +/- 0.135	0.683 +/- 0.041

Paired difference in Task A retention vs persistent (mean, exploratory 95% percentile bootstrap interval):

- `passive_002`: -0.108 ([-0.158, -0.057]; n=5).
- `passive_008`: -0.372 ([-0.400, -0.345]; n=5).
- `reinforced_002_4`: -0.027 ([-0.057, +0.006]; n=5).
- `reinforced_002_8`: -0.016 ([-0.042, +0.015]; n=5).
- `reinforced_008_4`: -0.116 ([-0.174, -0.060]; n=5).
- `reinforced_008_8`: -0.071 ([-0.127, -0.017]; n=5).



Interpretation and limitations

This study compares *descriptive held-out accuracy* and paired differences on small, synthetic domain changes. A larger Task A score together with a lower Task B score would imply a stability-plasticity trade-off, not necessarily a better memory algorithm. Do not describe any observed advantage as a verified continual-learning breakthrough or as statistically significant without a separately designed multiple-comparison-controlled study.

- The input is scikit-learn 8x8 digits, **not MNIST, Fashion-MNIST or CIFAR**.

- All ten classes are present in both domains; this differs from the Stage One disjoint-label task.

- The study does not include rehearsal/replay, EWC, task-order reversal, real-world concept drift or a larger neural architecture.

- Bootstrap intervals treat the seed as the paired unit and are **unadjusted for multiple comparisons**.

- Metrics measure classification accuracy, not human-like memory, physical energy, or neural data retention.

- A five-seed exploratory result should not be treated as a publication-quality claim. The next study needs held-out evaluation seeds, more baselines, ablations, and a prespecified primary comparison.

Reproduce and inspect the full evidence

- Stage Two protocol

- Frozen Stage One results

- Stage Two run CSVs, manifest and figures

Run: `python -m adaptive_memory.stage_two --output results/stage-two`. The linked Stage Two run is a versioned exploratory evidence bundle; do not modify it in place after publication.